AUTONOMOUS, RESPONSIBLE AI

Designing Geldium's Next-Generation Collections System

From risk insight to scalable, fair, and explainable automation

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From Insight to Action

Where this strategy builds from

Task 1 — EDA Findings

Credit Utilization, Debt-to-Income Ratio, and a worsening 6-month payment trend are the clearest early signals of delinquency risk.

Task 2 — Model Plan

A Gradient Boosted Decision Tree model, optimized for Recall, uses Missed_Payments, Credit Utilization, Payment History, Debt-to-Income Ratio, and Credit Score to flag risk.

Task 3 — Strategy

A First-Lapse Intervention program engages customers the moment they slip from On-time to Late — targeting a 15% cut in progression to full delinquency.

Today's task: design the system that operationalizes these insights at scale — autonomously, fairly, and within regulatory guardrails.

How the System Works

End-to-end flow: data in, action out, continuously learning

1. Customer Data

Credit utilization, debt-to-income, missed payments, 6-month payment history feed in daily from core banking + CRM systems.

2. Decision Logic

The GBDT risk model scores each account and assigns a risk tier (Low / Medium / High) using the Task 2 feature set.

3. Action

Risk tier triggers a matched intervention — automated nudge, supportive outreach, or escalation to a human agent.

4. Learning Loop

Outcomes (cured, escalated, defaulted) feed back into the model for retraining and drift monitoring.

feedback loop: outcomes from Stage 4 retrain the model feeding Stage 1

How the System Works

Inside the decision logic: model, inputs, and risk tiers

Gradient Boosted Decision Trees

Captures non-linear interactions (e.g. Income x Debt-to-Income) that simpler models like Logistic Regression miss — at the cost of needing SHAP for explainability.

TOP 5 INPUT FEATURES

- Missed Payments (12-mo count)
- Credit Utilization
- 6-Month Payment History trend
- Debt-to-Income Ratio
- Credit Score

RISK TIER OUTPUT

Low Risk

Stable payment history, low utilization — no action required, continues standard monitoring.

Medium Risk

First lapse from On-time to Late detected — triggers the automated First-Lapse Intervention (Task 3).

High Risk

Multiple missed payments + rising Debt-to-Income — escalated to a human collections specialist.

Role of Agentic AI

What runs autonomously vs. what stays human-in-the-loop

AUTONOMOUS	HUMAN OVERSIGHT
Daily risk scoring of all active accounts	Approval of high-risk escalations & legal action
Tier assignment (Low / Medium / High)	Review of borderline / contested risk scores
Automated SMS/email outreach for first-lapse cases	Sign-off on payment-plan or hardship offers
Routine model monitoring & drift alerts	Quarterly fairness & bias audit review

Responsible AI Guardrails

Fairness, explainability, and compliance built into the system

Fairness & Bias Monitoring

Demographic Parity and Disparate Impact tests run on every model refresh, checking that risk scores stay independent of Age and Location (per Task 2's bias detection plan).

Explainability (SHAP)

Every risk score is paired with a SHAP-based feature breakdown, so any customer or regulator can see exactly which factors — e.g. utilization, missed payments — drove the rating.

Regulatory Compliance

Outreach scripts and escalation paths are reviewed against fair-lending and debt-collection regulations before deployment; legal sign-off required for any new intervention type.

Human-Centered Design

Interventions are framed as support, not punishment — flexible payment plans are offered before any escalation, consistent with Task 3's responsible-AI recommendation.

Expected Business Impact

Quantitative KPIs and qualitative outcomes

BUSINESS KPIs

15%	reduction in Late-to-Delinquent progression (Task 3 pilot target, 90 days)
↓ Write-offs	via earlier, lower-cost intervention vs. late-stage collections
↑ Recall	on true delinquency cases — fewer missed at-risk accounts
↓ Manual Load	on collections staff — automation handles low/medium tiers

CUSTOMER OUTCOMES

- Earlier, supportive outreach instead of late-stage pressure tactics
- Consistent, explainable treatment — every flagged customer can see why
- Fairer outcomes across age and location, monitored every model cycle
- Faster access to flexible payment plans before accounts escalate

Recommendation

Pilot the autonomous collections system alongside the existing First-Lapse Intervention program for 90 days — autonomous scoring and low/medium-tier outreach live from day one, with full human review on every high-risk escalation.

Next steps: Legal/compliance sign-off → Fairness baseline audit → 90-day pilot launch → Quarterly review